

FDS LAB DAY 1 WORKOUT QUESTIONS

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1. Create a DataFrame Question: Create a DataFrame from the following dictionary:
python CopyEdit data = { 'Name': ['Alice', 'Bob', 'Charlie', 'David'], 'Age': [25, 30, 35, 40],
'City': ['New York', 'Los Angeles', 'Chicago', 'Houston'] } Display the DataFrame.

[2]

import pandas as pd

[3]
✓ Os

data = {
 'Name' : ['Alice', 'Bob', 'Charlie', 'David'],
 'Age' : [25, 30, 35, 40],
 'City' : ['New York', 'Los Angeles', 'Chicago', 'Houston'],
}
df = pd.DataFrame(data)

[4]
✓ Os

df

...

	Name	Age	City
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Charlie	35	Chicago
3	David	40	Houston

Next steps: [Generate code with df](#) [New interactive sheet](#)

2. Filtering Rows Question: From the above DataFrame, filter out the rows where the age is greater than 30

```
[5]  
✓ Os  
df[df['Age']>=30]
```

	Name	Age	City
1	Bob	30	Los Angeles
2	Charlie	35	Chicago
3	David	40	Houston

3. 3.Adding a New Column Question: Add a new column Salary with values [50000, 60000, 70000, 80000] to the existing DataFrame

```
df['Salary'] = [50000,60000,70000,80000]
```

df

	Name	Age	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Los Angeles	60000
2	Charlie	35	Chicago	70000
3	David	40	Houston	80000

4. 4. Update Column Values Question: Increase the salary of all people by 10%.

```
df['Salary'] = df['Salary'] * 1.10
```

df

	Name	Year	City	Salary
0	Alice	25	New York	55000.0
1	Bob	30	Los Angeles	66000.0
2	Charlie	35	Chicago	77000.0
3	David	40	Houston	88000.0

5. 5. Drop a Column Question: Remove the City column from the DataFrame

```
df.drop("City",axis=1)
```

...

	Name	Age	Salary
0	Alice	25	50000
1	Bob	30	60000
2	Charlie	35	70000
3	David	40	80000

6. 6. Rename Columns Question: Rename the column Age to Years.

```
df.rename(columns = {"Age":"Year"},inplace=True)
```

df

...

	Name	Year	City	Salary
0	Alice	25	New York	50000
1	Bob	30	Los Angeles	60000
2	Charlie	35	Chicago	70000
3	David	40	Houston	80000

7. 7. Sorting Values Question: Sort the DataFrame by Salary in descending order

```
df.sort_values(by="Salary", ascending=False)
```

	Name	Year	City	Salary
3	David	40	Houston	88000.0
2	Charlie	35	Chicago	77000.0
1	Bob	30	Los Angeles	66000.0
0	Alice	25	New York	55000.0

8. 8. Grouping and Aggregation Question: Create a new DataFrame: python CopyEdit data = { 'Department': ['HR', 'IT', 'HR', 'IT'], 'Employee': ['Alice', 'Bob', 'Charlie', 'David'], 'Salary': [50000, 60000, 55000, 65000] } Group by department and find the average salary.

```
import pandas as pd

data = {
    'Department': ['HR', 'IT', 'HR', 'IT'],
    'Employee': ['Alice', 'Bob', 'Charlie', 'David'],
    'Salary': [50000, 60000, 55000, 65000]
}

df = pd.DataFrame(data)

print(df)
```

```
***      Department Employee  Salary
0         HR      Alice   50000
1         IT       Bob    60000
2         HR   Charlie   55000
3         IT      David   65000
```

9. 9. Merge Two DataFrames Question: Given two DataFrames: python CopyEdit df1 = pd.DataFrame({ 'ID': [1, 2, 3], 'Name': ['Alice', 'Bob', 'Charlie'] }) df2 = pd.DataFrame({ 'ID': [1, 2, 4], 'Score': [85, 90, 88] }) Merge them on the ID column.

```
import pandas as pd

df1 = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 4],
    'Score': [85, 90, 88]
})

result = pd.merge(df1, df2, on='ID')

print(result)
```

```
***      ID  Name  Score
0       1  Alice     85
1       2   Bob     90
```

10. 10. Handling Missing Values Question: Create a DataFrame with some missing values and fill the missing data with appropriate default values.

```
import pandas as pd

df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'Age': [20, None, 22, None],
    'Salary': [50000, 60000, None, 70000]
})

print("Before filling missing values:")
print(df)

df['Age'] = df['Age'].fillna(0)
df['Salary'] = df['Salary'].fillna(0)

print("\nAfter filling missing values:")
print(df)
```

... Before filling missing values:

	Name	Age	Salary
0	Alice	20.0	50000.0
1	Bob	NaN	60000.0
2	Charlie	22.0	NaN
3	David	NaN	70000.0

After filling missing values:

	Name	Age	Salary
0	Alice	20.0	50000.0
1	Bob	0.0	60000.0
2	Charlie	22.0	0.0
3	David	0.0	70000.0